SSc-MIM

A curated, SBGN-compliant Molecular Interaction Map of skin fibrosis in diffuse cutaneous systemic sclerosis

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Expert biology review session · June 2026

github.com/Nurtal/IDT_SSc_map

Systemic sclerosis lacks a mechanistic map

The disease

Systemic sclerosis (scleroderma) — autoimmune fibro-inflammatory disease: progressive skin & organ fibrosis, vasculopathy, immune dysregulation.

Highest mortality of any rheumatic disease — and no disease-modifying therapy is approved.

The gap

Curated disease maps exist for rheumatoid arthritis (RA-map), Sjögren, COVID-19, Parkinson…

None exists for SSc. Its core molecular circuits are scattered across the literature, never assembled into one model.

Build the first SSc disease map, focused on dcSSc skin fibrosis.

Standards-based, fully sourced, citable (GitHub + Zenodo DOI).

Layered with real patient single-cell data — an analysis substrate, not just a diagram.

CONCEPT

What is a Molecular Interaction Map?

A computable, standards-based diagram of the molecular events driving a disease — species (genes, proteins, complexes, phenotypes) connected by mechanistic reactions.

SBGN

Systems Biology Graphical Notation — an unambiguous visual grammar for biology

CellDesigner

The editor the map is built in; exports standard SBML (machine-readable)

MI2CAST

Annotation standard: every reaction carries its evidence, source PMID & ECO code

Reproducible

Public repo, scripted pipeline, Docker, continuous-integration validation

OVERVIEW

The SSc-MIM at a glance

568
molecular species

308
mechanistic reactions

20
cell & tissue compartments

4
biological modules

133
hand-curated SSc-specific reactions

~360
PubMed references mined & filled

197
patients (121 SSc / 76 HC) across 4 single-cell datasets

Built in CellDesigner, validated as SBML (0 errors in CI), annotated to MI2CAST, and released openly. Topology comes from the literature; patient data is layered on top to ground and read out the map.

How the map was built

Topology is literature-derived and fully sourced — it is not inferred from the omics data.

1

Scope

Define 4 modules + 4 phenotype sinks; Tier-1 entity tables set the boundaries.

→

2

Backbone import

Canonical pathways from Reactome (IFN-I, TGF-β/PDGF, NOTCH1, IL-6); harmonised & de-duplicated across modules.

→

3

SSc curation

133 SSc-specific reactions hand-added from a 77-paper worklist + co-author full-text PDFs; each MI2CAST-annotated.

→

4

Assemble

Wire everything in CellDesigner into one SBML map: 568 species / 308 reactions / 20 compartments.

→

5

QC gates

SBML validation in CI, grounding gate (no unsourced edges), contradiction detector, AI verdict + citation sweep.

A computable, fully-sourced map where every edge is traceable to a verified reference — ready for analysis and expert review.

Four biologically coherent modules

M1 — Type-I interferon

cGAS–STING, RIG-I/MAVS, TLR3/7/9 → IRF3/7 → ISG signature. The innate-immune ignition of SSc.

M2 — TGF-β & fibrosis

Latent TGF-β activation → SMAD2/3 → fibroblast-to-myofibroblast transition, collagen & ECM. The fibrotic core.

M3 — EndoMT & vasculopathy

Endothelin, Notch, NO/sGC, HIF → endothelial-to-mesenchymal transition & vascular remodelling.

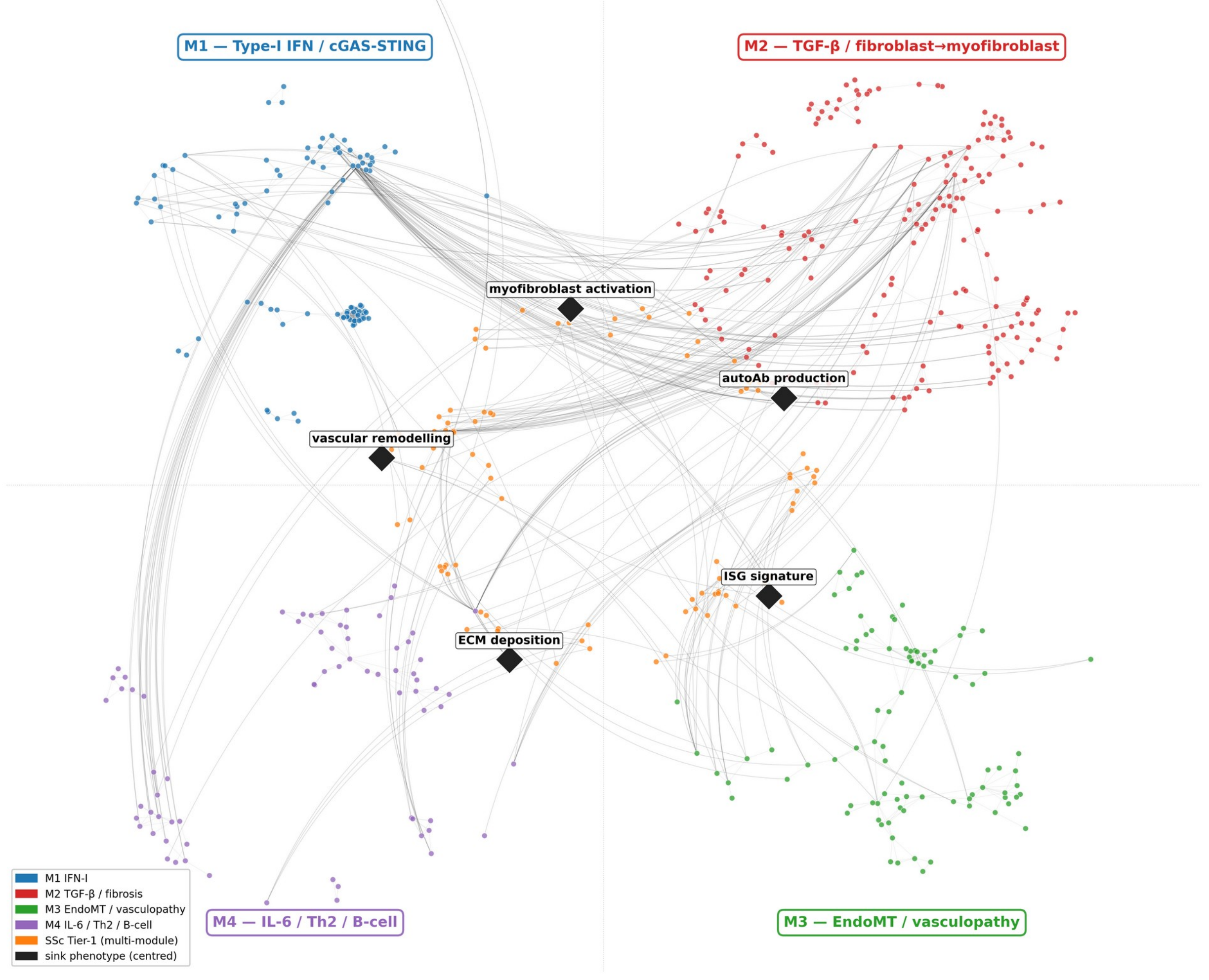
M4 — IL-6 / Th2 / B cells

IL-6/STAT3, IL-4/IL-13/STAT6, BAFF–BCMA, autoantibodies. The adaptive-immune & cytokine layer.

Plus a crosstalk layer wiring the modules together (e.g. IL-6 → SMAD3, IFN-I → fibroblast).

One integrated, modular network

Figure 1 — SSc-MIM global view (quadrant layout)
526 species · 1488 edges · modules in fixed positions · phenotype sinks centred · inter-module edges shown as curved arcs



Each quadrant is one module; nodes are species, edges are mechanistic reactions.

Hubs that emerge from network analysis match known SSc biology: TGF-β:receptor, SMAD3–SMAD4, ISG signature, PDGFR.

The same SBML file drives the figure, the analysis, and the online viewer.

From molecules to disease phenotypes

The map is built around four phenotype “sinks” — the biological outputs of the disease. Modelling rule: every disease-relevant entity must reach a sink in ≤ 6 steps (verified: 0 violations).

Myofibroblast activation

ECM / collagen deposition

Vascular remodelling

ISG / interferon signature

It turns a static drawing into a testable model: a perturbation anywhere has a traceable path to a clinical readout.

It makes the scope explicit and reviewable — exactly what we want expert biologists to challenge.

Every reaction is sourced and graded

Tiered citation policy (GO-aligned)

Each reaction carries a primary PubMed ID and an ECO evidence code — experimental evidence ranked above inference.

Provenance is tracked

Every edge records where it came from: Reactome import, hand curation, or literature mining — fully auditable.

No fabricated citations — ever

A built-in grounding gate rejects any claim without a real, verifiable source (a negative-control fixture proves it works).

~360 references mined & auto-filled (title, authors, journal) via NCBI E-utils.

133 SSc-specific reactions curated from a systematic 77-paper worklist + co-author full-text PDFs.

MI2CAST-compliant annotation throughout.

Four public single-cell datasets

197 donors (121 SSc / 76 HC) across skin, lung and blood — all re-analysed from raw GEO files.

GEO accession	Reference (verified)	Tissue	Donors (SSc/HC)
GSE195452	Gur C et al. Cell 2022;185:1373	Skin (multiome)	154 (97 / 57)
GSE138669	Tabib T et al. Nat Commun 2021;12:4384	Skin	22 (12 / 10)
GSE128169	Morse C et al. Eur Respir J 2019;54:1802441	Lung (ILD)	13 (8 / 5)
GSE210395	GEO — no linked publication	PBMC	8 (4 / 4)

Skin cohorts (Gur, Tabib) give fibroblast-subset resolution — the actors M2/M3 model; ≈100k and 64k annotated cells.
Lung (SSc-ILD) and PBMC extend the map beyond skin. Every citation above was re-checked against PubMed.

How patient data is layered on the map

1

Re-process

scanpy QC + cell-type annotation of each dataset from raw counts.

→

2

Pseudobulk DEG

Per-donor pseudobulk; SSc-vs-HC differential expression (mixed-effects NB GLM, BH-FDR).

→

3

Map onto species

Match differentially-expressed genes to map species → coverage.

→

4

Score per donor

AUCell turns each patient into a 4-module activation vector.

81.3%

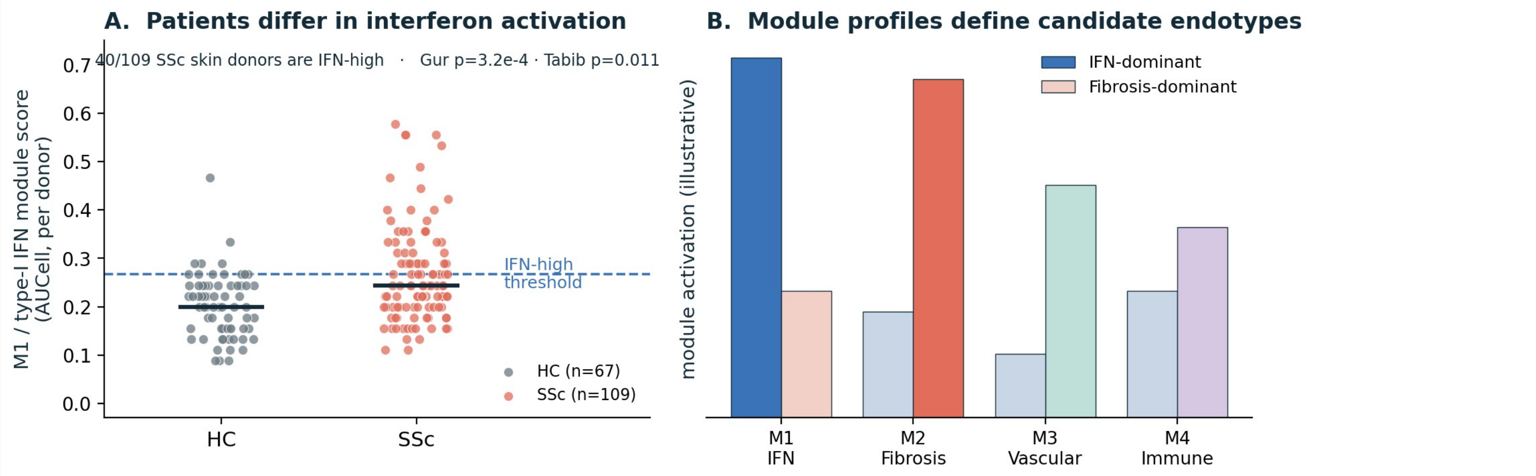
of detectable map species observed in patient data

M1 84% · M2 88% · M3 75% · M4 71% — the curated biology is largely measurable in real patients.

Biological check: M1 / type-I IFN module significantly elevated in SSc skin (Gur $p = 3.2\times10^{-4}$; Tabib $p = 0.011$).

60 ready-to-load MINERVA overlays for interactive exploration.

Toward endotypes: distinct activation profiles



Each patient → a 4-module activation vector. Distinct profiles → mechanism-matched endotypes (IFN-high → JAK/IFN axis · fibrosis-high → TGF-β/CTGF axis), recapitulating known SSc intrinsic subsets.

The map converts each patient into a 4-module activation vector.

Real result: IFN activation is heterogeneous — 40/109 SSc skin donors are IFN-high, others are not (reproducible across Gur + Tabib).

Distinct module profiles → candidate, mechanism-matched endotypes (IFN-dominant vs fibrosis-dominant).

Recapitulates the known SSc intrinsic subsets (inflammatory ↔ M1/M4, fibroproliferative ↔ M2).

Clinical promise: module-matched therapy — IFN-high → JAK/IFN axis; fibrosis-high → TGF-β/CTGF axis.

Bring your own patient clusters

Have 3 patient clusters? Project them onto the map and read out what mechanistically separates them — same pipeline, your labels.

YOUR CLUSTERS

- Cluster 1
- Cluster 2
- Cluster 3

→

AUCell — score each cluster on the 4 modules (M1/M2/M3/M4 + Tier-1).

Per-cluster differential expression mapped onto map species → which reactions light up.

MINERVA overlay — colour the map by each cluster (any cluster label, no new method).

WHAT YOU GET BACK — per cluster

Module fingerprint

A 4-module activation vector per cluster — e.g. cluster 2 IFN-high, cluster 3 fibrosis-dominant.

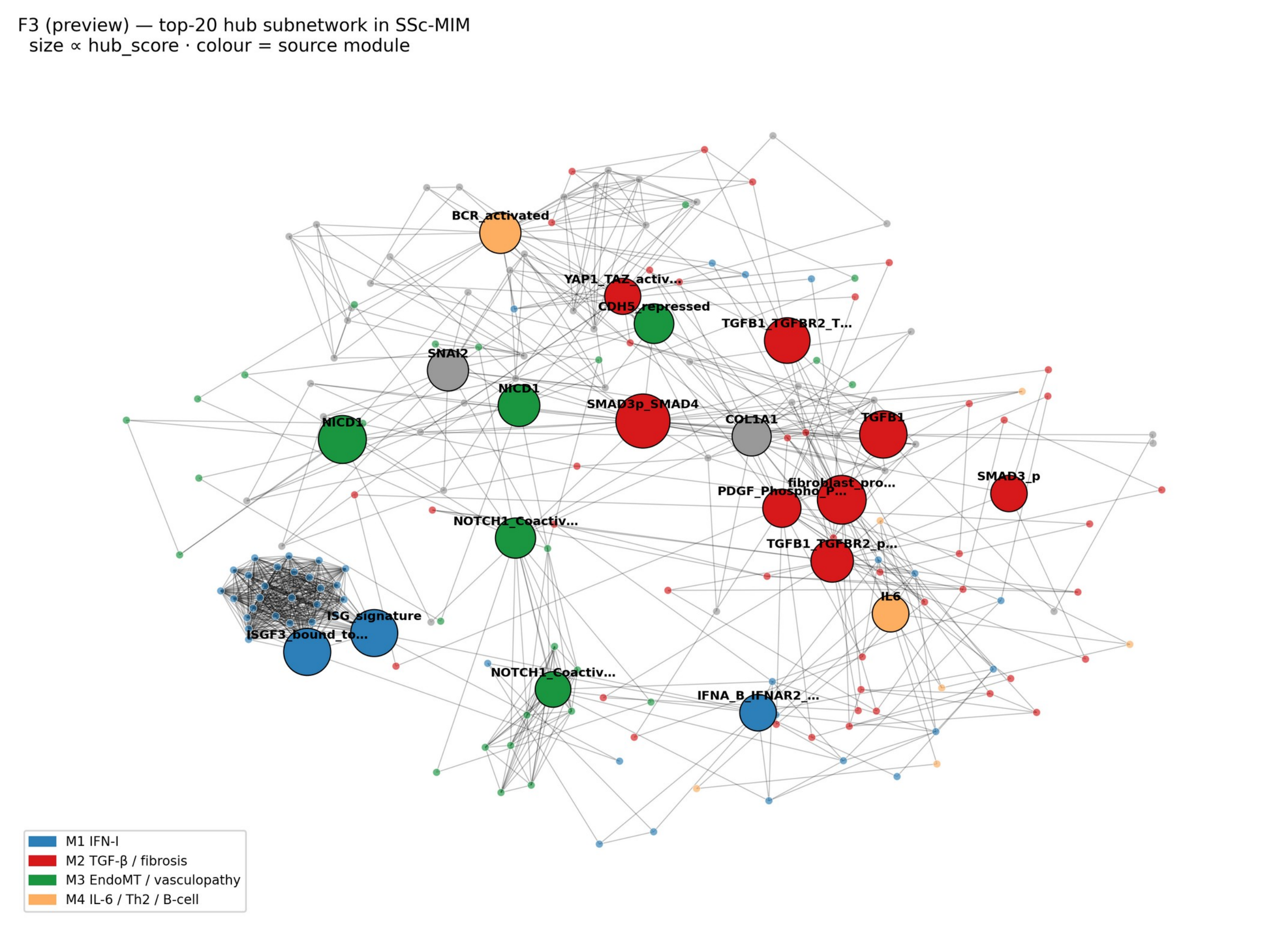
Mechanistic hypotheses

The lit-up sub-circuits that distinguish each cluster — e.g. STING→IRF3 vs TGF-β→SMAD3→collagen.

Candidate targets

Active druggable hubs per cluster → a cluster-matched therapy hypothesis to test.

A substrate for drug prioritisation



Network hubs cross-referenced with DGIdb drug–target data.

Recovers targets already in SSc trials — tocilizumab (IL-6), nintedanib, romilkimab (IL-13), pamrevlumab (CTGF/CCN2), bosentan (endothelin).

Drug table recalibrated against real SSc clinical-trial outcomes.

Shows the map can rank and rationalise repurposing candidates.

Independent verification of every interaction

143
interactions queued for expert review

128
passed automated AI verification

28
wrong source PMIDs caught & fixed

0
fabricated citations

Contradiction detector

Flags any gene pair given opposite signs by different sources, for human arbitration.

AI reviewer pass

Each interaction checked against its deciding sentence + supporting/contrary abstracts — verdict: validate / revise / caution.

Citation-integrity sweep (this week)

Found 28 reactions whose cited PMID pointed to an unrelated paper (biology canonical, reference a data-entry slip — e.g. a TGF-β reaction cited to a rubella study). Every replacement PMID was re-verified against live PubMed before applying.

An app to review the map, one interaction at a time

A swipe deck for curation

Offline, self-contained web app — one card per interaction, accept / reject / annotate.

Everything you need to decide, on the card

Regulator → target, mechanism, SSc relevance, the verbatim deciding sentence from the source article, and PubMed/DOI links.

Context before you judge

A literature dossier lists candidate supporting and “possibly contrary” references; an in-browser module map shows the neighbourhood.

Advisory AI verdict

Each card shows the AI call + rationale — filter to triage, but the human reviewer always decides.

No install, no server — open one HTML file.

Decisions saved locally; export to CSV/JSON.

Re-includes even discarded edges, so you keep full control.

Demo right after this.

WHAT WE ASK OF YOU

Three things from expert biologists

1

Validate the scope

Do the 4 modules + phenotype sinks capture dcSSc skin fibrosis? What is missing or over-reaching?

2

Curate the entities

Tier-1 SSc species: what to add, remove, or promote — with PMIDs where you have them.

3

Adjudicate interactions

Use the review app on the 143 interactions: confirm, reject, or flag for revision.

Where this is going

An open, citable SSc resource — and your review is the gate

v1.0 release: GitHub + Zenodo DOI (citable).

Resource paper (npj Systems Biology & Applications) — revision drafted.

Endotype stratification as the next translational step.

Expert sign-off on scope & interactions is the last binding step.

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github.com/Nurtal/IDT_SSc_map

Open the review app now for a live walkthrough.

Building the SSc-MIM

Construction approach, datasets, and the interaction-validation process

Technical companion to the SSc-MIM overview

Nathan Foulquier — LBAI, UMR 1227 Inserm · CHU de Brest

Detailed reference: docs/SSc_MIM_construction_and_validation.md

Three governing principles

1. Deterministic curation

Explicit rules (Mazein 2023 + Disease Maps conventions) so two curators produce the same map. HGNC names, fixed compartments, fixed reaction granularity.

2. Never written speculatively

No edge enters the curated map without a verbatim source quote that passes automated gates AND human ratification.

3. Topology is curated, not learned

Mechanisms come from the literature; patient omics ground, validate and read out the map — they never define its structure.

Notation, format & naming conventions

Notation & format

CellDesigner v4.4 · SBGN Process Description glyphs.

SBML Level 2 Version 4 + CellDesigner annotations.

MI2CAST-compliant annotation on every reaction.

Identifiers (HGNC-anchored)

```
gene/protein  → HGNC symbol  (TGFB1, SMAD3)
proteoform    → symbol+state (SMAD3@S423)
complex       → A:B:C  (SMAD2:SMAD3:SMAD4)
phenotype     → phenotype_<slug>
```

One reaction per ligand–receptor binding event.

One reaction per kinase–substrate pair (residues collapsed into a state-variable list).

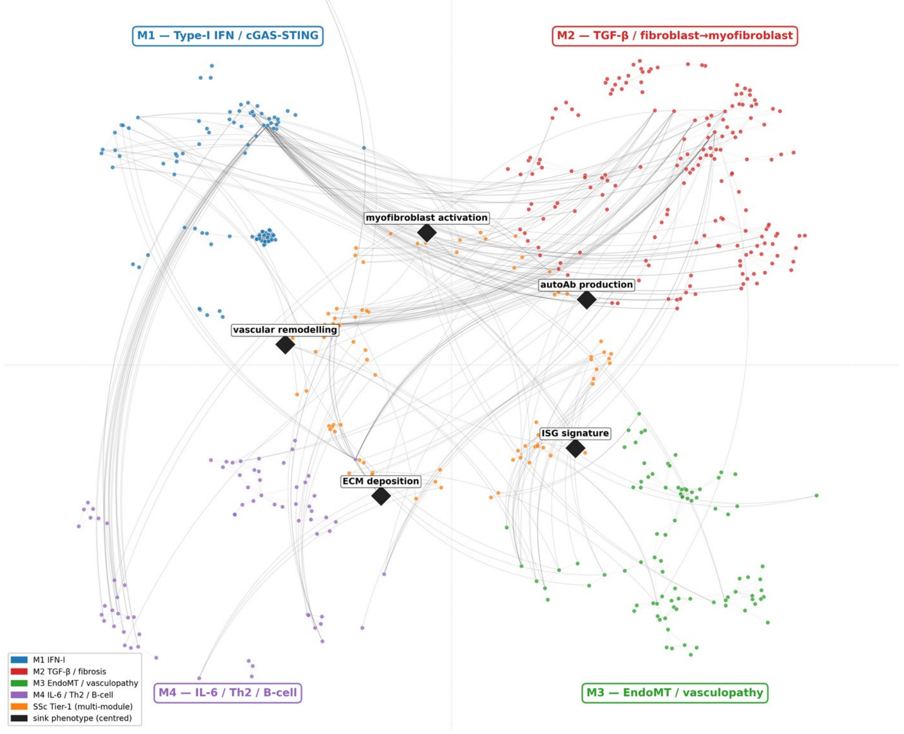
Transcription as TF → mRNA(target), one per target.

Fixed compartment vocabulary (20): extracellular, ECM, plasma membrane, cytosol, nucleus, ER, endosome, mitochondrion, cell-type prefixes…

ARCHITECTURE

Modules, crosstalk & phenotype sinks

Figure 1 — SSc-MIM global view (quadrant layout)
526 species · 1488 edges · modules in fixed positions · phenotype sinks centred · inter-module edges shown as curved arcs



M1 Type-I IFN • 19 reactions

M2 TGF-β / fibrosis • 58

M3 EndoMT / vasculopathy • 27

M4 IL-6 / Th2 / B cells • 21

crosstalk • 8

Four phenotype sinks (disease outputs): myofibroblast activation · ECM/collagen deposition · vascular remodelling · ISG signature.

Modelling rule: every entity reaches a sink in ≤ 6 steps — verified, 0 violations (make sink-check).

PIPELINE

From literature to a released map

Blue = backbone assembly · coral = SSc-specific layer · teal = QC & release.

Each stage is a scripted, reversible step (make targets) with linters and CI — re-running from raw inputs reproduces the map.

The whole automated lane runs end-to-end with `make auto`.

STEP 1–4

Reactome backbone import & integration

Import canonical pathway scaffolds

M1 type-I IFN · M2 TGF-β (pilot R-HSA-2173789) + PDGF · M3 NOTCH1 (R-HSA-1980143) · M4 IL-6.

Post-process & harmonise

Decode + classify reaction types; rename entities to HGNC; de-duplicate species shared across modules.

Integrate

Merge harmonised module XMLs into one SBML map. Mine embedded PMIDs; auto-fill the bibliography via NCBI E-utils.

```
reactome_pilot.py       (pilot)
post_process_reactome   (harmonise)
harmonise_imports.py    (harmonise)
integrate_modules.py    (integrate)
extract_pmids_*.py      (pmids)
bib_lookup.py           (bib-lookup)
```

Backbone = 159 reactions; gives canonical structure, but propagates curator-inference ECO by default.

STEP 5

The SSc-specific curation layer — the novel content

133 hand-curated reactions Reactome cannot structurally hold (autoantibodies, SSc cell states, GWAS-function, clinical crosstalk). Sourced from a systematic 77-paper worklist (24 themes) + co-author full-text PDFs.

Provenance	n	Meaning
original-curation	40	hand-curated (human-original)
fulltext-verified	27	AI-extracted, verbatim-grounded
ai-discovery (gated)	48	via G0–G4 pipeline
claude-reclassify	10	cell-state → conceptual bridge
claude-lit-mine	8	mined, abstract-verified

```
M2  58   M3  27   M4  21
M1  19   crosstalk  8
```

Every AI-touched row is tagged, reversible (one TSV row), and carries its grounding quote in notes.

126 / 133 carry a primary PMID.

Provenance & graded evidence

The honest denominator: separate the imported backbone from the genuinely SSc-specific layer.

Layer	Rxns	PMID	Exp. ECO
Reactome backbone	159	99%	0
SSc-specific	133	95%	majority

ECO code	Evidence	n
0000314	direct assay	73
0000033	traceable author stmt	27
0000270	expression pattern	20
0000305	curator inference	9
others (353/315)	interaction / mutant	4

122 confirmed · 6 phenotype_aggregation (definitional sinks) · 4 conceptual_bridge · 1 to_complete (POSTN, under expert review).

make evidence-lint fails CI if any SSc reaction is left untriaged — no undeclared inference debt.

Adding an interaction: the edge-discovery protocol

Governing rule: the curated map is never written speculatively.

1. **Stage**
   Candidate edge in staging TSV — with a verbatim supporting sentence from the source paper.
2. **Fetch**
   Cache source text (Europe PMC OA full text / abstract).
3. **Gate**
   Run G0–G4 automated gates → validation_report.tsv.
4. **Ratify**
   A human sets decision = promote (default = held).
5. **Promote**
   Only PASS + promote rows enter the map — tagged, reversible.

Candidates never touch the curated TSV until promoted.

Quality over volume — 12 grounded edges beat 40 shaky ones.

Disease-specific only — content Reactome cannot hold.

One quote → one edge.

Every promoted edge is one removable TSV row.

Five automated gates (G0–G4)

Gate	Check	On failure
G0 schema	required fields present; type in controlled vocabulary	REJECT
G1 HGNC	every gene entity is an official HGNC symbol (HGNC REST)	REJECT (alias→FLAG)
G2 grounding	supporting quote is a verbatim substring of the source text	REJECT — keystone
G3 novelty	gene pair not already an SSc reaction; not a forward Reactome pair	REJECT (dup) / FLAG
G4 evidence	numeric PMID present; corpus text available (SSc context)	FLAG

If the quote is not literally in the paper, the edge is rejected — the core defence against fabricated biology.

Self-test: a fabricated negative-control edge (cand_negctrl, FOXP3→COL1A1, invented quote + decoy PMID) must REJECT on every run; it is hidden from the reviewer app.

Layered QC beyond the gates

Structural

SBML L2V4 validation in CI · sink-connectivity (≤6 steps) · CellDesigner-loadability · MINERVA preflight.

Contradiction detector

Flags any gene pair given opposite polarities by different sources → human arbitration. 0 unresolved.

AI verdict pass

Read each interaction's deciding quote + support/contrary abstracts → validate / revise / caution. Now 128 validate · 5 caution.

Citation-integrity sweep

Caught 28 reactions citing an unrelated PMID; every replacement re-verified vs live PubMed. No PMID cited from memory.

Reviewer app

Offline swipe-deck over 143 interactions: verbatim quote, dossier, module map, advisory verdict; decisions export to CSV/JSON.

Human ratification

Binding step: the expert/co-author adjudicates the biology. AI proposes & self-gates; the human decides.

Four public single-cell datasets

197 donors (121 SSc / 76 HC) — skin, lung, blood — re-analysed from raw GEO files. Citations verified against PubMed.

GEO	Reference (verified)	Tissue	Donors (SSc/HC)
GSE195452	Gur C et al. Cell 2022;185:1373	Skin (multiome)	154 (97 / 57)
GSE138669	Tabib T et al. Nat Commun 2021;12:4384	Skin	22 (12 / 10)
GSE128169	Morse C et al. Eur Respir J 2019;54:1802441	Lung (ILD)	13 (8 / 5)
GSE210395	GEO — no linked publication	PBMC	8 (4 / 4)

Skin cohorts (≈100k + 64k annotated cells) give the fibroblast-subset resolution that M2/M3 model.

Lung (SSc-ILD) + PBMC extend the map beyond skin. Raw archives SHA-256-pinned in data/MIRROR.md (Zenodo mirror).

How the datasets are used

1

Re-process

scanpy QC, Leiden clustering, cell-type annotation from raw counts.

2

Pseudobulk DEG

Per-donor pseudobulk; SSc-vs-HC mixed-effects NB GLM + BH-FDR (q=0.05).

3

Map onto species

Match DE genes to map species → coverage.

4

Score per donor

AUCell → each patient = a 4-module activation vector.

5

Overlay

60 MINERVA overlays colour the map by cluster/dataset.

81.3%

of detectable map species seen in patient data

Per-module coverage: M1 84% · M2 88% · M3 75% · M4 71% · Tier-1 83%.

M1 / type-I IFN significantly up in SSc skin (Gur p=3.2e-4; Tabib p=0.011).

Cell-type labels validated with CellTypist (κ=0.92, ARI=0.70). Data grounds the map — it does not define its topology.

Per-donor pseudobulk — the right unit of replication

Tens of thousands of cells, but only 4–13 patients per dataset. The biological question is about patients, not cells.

The problem: pseudoreplication

Cells from one patient are not independent (same genotype, biopsy, batch).

Testing per-cell treats them as thousands of samples → inflated power, many false positives (reviewer R2-M1).

The fix

Rebuild one bulk-like profile per patient → the unit of replication becomes the patient, the correct level for SSc-vs-HC.

```
group cells by (donor × cell_type)
row[gene] = Σ raw counts over cells
```

→ one pseudobulk count row per (donor, cell type).

Then DEG on the pseudobulks

```
NB GLM (pydeseq2) ~ condition
+ Benjamini-Hochberg FDR (q=0.05)
```

Raw counts (not log) — the NB GLM models depth & overdispersion itself.

Trade-off: fewer, more robust hits. Less cell-level granularity, but a valid, reproducible inference — and the same pseudobulks feed AUCell.

AUCell — label-free module activation scoring

"Are a module's genes concentrated among this patient's most highly-expressed genes?" If yes → the module is on.

Why — no double-dipping

v1.0 weighted expression by the SSc-vs-HC sign, then tested SSc-vs-HC → circular (reviewer R2-M2).

AUCell scores each donor against the pre-specified module gene set (from the curated map), blind to the SSc/HC label.

Rank-based → robust

Depends only on the order of genes, not absolute values → insensitive to scale, depth, normalisation.

Rank all genes by expression (rank 1 = highest).

Walk the top T (top ~5%); recovery curve R(r) = # module genes with rank ≤ r.

`AUC = Σ R(r),  normalised to [0,1]`

High AUC → module genes cluster at the top → module active.

Output: a 4-module activation vector per patient (+ a Z-score for triangulation).

Feeds the SSc-vs-HC contrasts (M1 IFN p=3.2e-4 in skin) and the endotype workflow (per-cluster module fingerprints).

Scripted, tested, containerised, citable

Scripted pipeline

make auto runs the whole lane; make lint / validate run every content + SBML linter.

Tested + CI

pytest suite (no data needed); GitHub Actions: validate_sbml, lint, scripts-smoke, figures.

Containerised

Dockerfile + workflow pin the environment for end-to-end reproduction.

Provenance & metadata

RO-Crate manifest, MIRIAM injection for BioModels, CITATION.cff + .zenodo.json.

Input-data mirror

data/MIRROR.md + .sha256 — SHA-256-pinned GEO archives (Zenodo mirror).

Release

GitHub + Zenodo DOI at git tag v1.0; make release pre-flights.

In one sentence

A literature-curated map, gated against fabrication, grounded in real patient data

Backbone from Reactome → harmonised → integrated into one SBML map (568 sp / 308 rx / 20 comp).

133 SSc-specific reactions added only through G0–G4 gates + human ratification — no edge without a verbatim source.

Four single-cell datasets (197 donors) ground the map: 81.3% coverage, type-I IFN validated in SSc skin.

Fully scripted, tested, containerised and citable (Zenodo DOI, BioModels-ready).

SSc-MIM

Building the map, validating it, and characterising SSc endotypes

Construction · validation (gates + patient data) · endotype use case

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The map in one screen

568
molecular species

308
mechanistic reactions

133
SSc-specific reactions

197
patient donors

A curated, SBGN/SBML disease map for dcSSc skin fibrosis.

Four modules — M1 type-I IFN · M2 TGF-β/fibrosis · M3 EndoMT/vasculopathy · M4 IL-6/Th2/B — draining into 4 phenotype sinks.

Topology is literature-curated; patient single-cell data is layered on to validate it and read it out.

This talk: how it is built, how it is validated (two ways), and how it can characterise patient endotypes.

PART I

Building the map

Literature-curated topology, assembled and annotated — not inferred from omics.

From literature to an integrated map

Scope → Reactome import → Harmonise & dedup → Integrate → SSc curation (gated) → Annotate PMID/ECO → QC

Backbone (blue)

Canonical pathways imported from Reactome (IFN-I, TGF-β/PDGF, NOTCH1, IL-6), harmonised to HGNC, de-duplicated, merged into one SBML map.

SSc-specific layer (coral)

133 hand-curated reactions Reactome cannot hold (autoantibodies, SSc cell states, crosstalk) — added only through gated discovery (Part II).

```
M1 IFN 19   M2 fibrosis 58
M3 EndoMT 27   M4 immune 21   xt 8
```

4 phenotype sinks: myofibroblast · ECM · vascular remodelling · ISG signature.

Rule: every entity reaches a sink in ≤6 steps (0 violations).

PART II

Validating the map

Two complementary axes: (A) gated against fabrication, (B) grounded in real patient data.

Two complementary axes

A • Is every edge real?

Edge-discovery protocol: nothing enters the curated map speculatively.

Five automated gates (G0–G4), G2 = verbatim-quote grounding.

Contradiction detector + advisory AI verdicts + citation-integrity sweep.

Binding human ratification by the expert/co-author.

B • Does it hold in patients?

Four single-cell datasets (197 donors) re-analysed from raw counts.

Per-donor pseudobulk → mixed-effects DEG → map coverage.

AUCell → per-patient module activation profile.

Result: 81.3% coverage; type-I IFN validated in SSc skin.

Gate strategy — the edge-discovery protocol

Governing rule: the curated map is never written speculatively.

1. **Stage** — candidate edge in a staging file — with a verbatim sentence from the source paper
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3. **Gate** — run the five G0–G4 gates → PASS / FLAG / REJECT
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5. **Promote** — only PASS + promote enter the map — tagged, reversible

The five gates (G0–G4)

Gate	Check	On failure
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If the quote is not literally in the paper, the edge is rejected.

Self-test: a fabricated negative-control edge (FOXP3→COL1A1, invented quote + decoy PMID) must REJECT on every run.

Then: contradiction detector · AI verdicts (128 validate / 5 caution) · citation sweep (28 wrong PMIDs fixed, re-verified vs PubMed) · human ratification.

Grounding in patient data — the datasets

197 donors (121 SSc / 76 HC) across skin, lung and blood — re-analysed from raw GEO files; citations verified on PubMed.

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GSE138669	Tabib T et al. Nat Commun 2021;12:4384	Skin	22 (12 / 10)	64 211
GSE128169	Morse C et al. Eur Respir J 2019;54:1802441	Lung (ILD)	13 (8 / 5)	67 516
GSE210395	GEO — no linked publication	PBMC	8 (4 / 4)	34 619

Skin cohorts give the fibroblast-subset resolution that M2/M3 model; lung (SSc-ILD) + PBMC extend beyond skin.

Raw archives SHA-256-pinned in data/MIRROR.md for a reproducible input envelope (Zenodo mirror).

Per-donor pseudobulk — the right unit of replication

Tens of thousands of cells, but only 4–13 patients per dataset. The question is about patients, not cells.

The problem: pseudoreplication

Cells from one patient are not independent (same genotype, biopsy, batch).

Testing per-cell treats them as thousands of samples → inflated power, false positives.

The fix

Rebuild one bulk-like profile per patient → the unit of replication becomes the patient.

```
group cells by (donor × cell_type)
row[gene] = Σ raw counts over cells
```

→ one pseudobulk count row per (donor, cell type).

```
NB GLM (pydeseq2) ~ condition
+ Benjamini-Hochberg FDR (q=0.05)
```

Raw counts (not log) — the NB GLM models depth & overdispersion itself.

Trade-off: fewer but robust hits; a valid, reproducible inference. The same pseudobulks feed AUCell.

AUCell — label-free module activation scoring

"Are a module's genes concentrated among this patient's most highly-expressed genes?" If yes → the module is on.

Why — no double-dipping

Scores each donor against the pre-specified module gene set (from the curated map), blind to the SSc/HC label.

Avoids the circularity of weighting by the same SSc-vs-HC contrast you then test.

Rank-based → robust

Depends only on gene order, not absolute values → insensitive to scale, depth, normalisation.

Rank all genes by expression (rank 1 = highest).

Walk the top T (top ~5%); recovery curve R(r) = # module genes with rank ≤ r.

`AUC = Σ R(r),  normalised to [0,1]`

High AUC → module genes cluster at the top → module active.

Output: a 4-module activation vector per patient.

This per-patient vector is exactly what powers the endotype use case (Part III).

What the data shows

81.3%

of detectable map species seen in patient data

Coverage by module

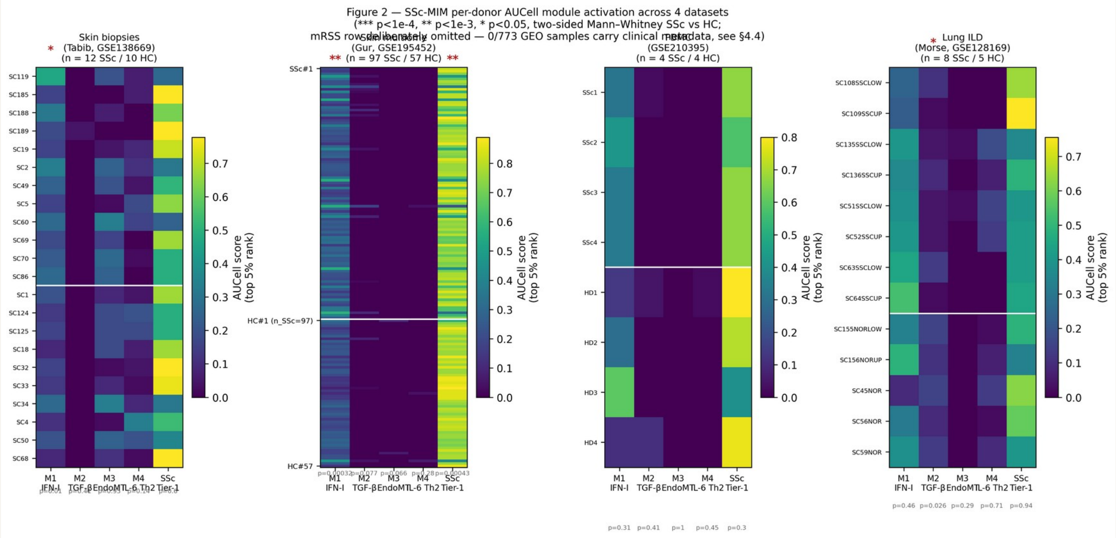
M1 84% · M2 88% · M3 75% · M4 71% — the curated biology is largely measurable.

Biological validation

Type-I IFN (M1) significantly elevated in SSc skin: Gur p=3.2e-4, Tabib p=0.011.

Annotation validated

CellTypist agreement κ=0.92, ARI=0.70 vs marker labels.



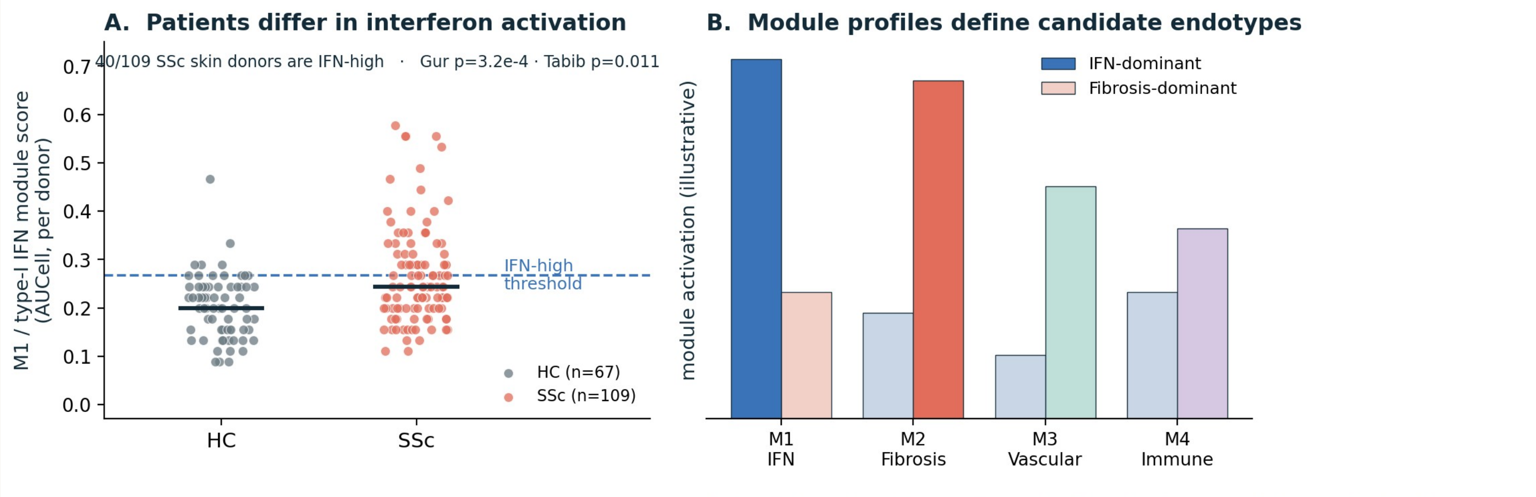
Per-donor AUCell module scores, SSc vs HC.

PART III

Toward endotypes

Using the validated map to characterise patient subgroups — mechanistically.

Patients already differ in module activation



Each patient → a 4-module activation vector. Distinct profiles → mechanism-matched endotypes (IFN-high → JAK/IFN axis · fibrosis-high → TGF-β/CTGF axis), recapitulating known SSc intrinsic subsets.

Each patient = a 4-module activation vector (Part II-B).

IFN activation is not uniform: 40/109 SSc skin donors are IFN-high, others are not (reproducible Gur + Tabib).

Distinct module profiles → candidate mechanism-matched endotypes (IFN-dominant vs fibrosis-dominant).

Recapitulates the known SSc intrinsic subsets (inflammatory ↔ M1/M4, fibroproliferative ↔ M2).

USE CASE

You arrive with 3 patient clusters — what the map gives you

Project any patient grouping onto the map — same pipeline (AUCell + per-cluster DEG + MINERVA overlay), your labels.

YOUR CLUSTERS

Cluster 1

Cluster 2

Cluster 3

→

AUCell — score each cluster on the 4 modules (M1/M2/M3/M4 + Tier-1).

Per-cluster differential expression mapped onto map species → which reactions light up.

MINERVA overlay — colour the map by each cluster (any label, no new method).

WHAT YOU GET BACK — per cluster

Module fingerprint

a 4-module activation vector — e.g. cluster 2 IFN-high, cluster 3 fibrosis-dominant.

Mechanistic hypotheses

the lit-up sub-circuits that distinguish each cluster — e.g. STING→IRF3 vs TGF-β→SMAD3→collagen.

Candidate targets

active druggable hubs per cluster → a cluster-matched therapy hypothesis to test.

In one line

A curated map, gated against fabrication, validated in patients — and ready to type them

Literature backbone + 133 gated SSc-specific reactions → one SBML map (568 sp / 308 rx).

G0–G4 + G2 grounding + human ratification: no edge without a verbatim source.

4 datasets, 197 donors; per-donor pseudobulk + AUCell; 81.3% coverage; IFN validated in SSc skin.

Project your patient clusters → per-cluster module fingerprint, mechanistic hypotheses, candidate targets.